

# Galois orbits & minimal polynomials

Proposition ((5)  $\Rightarrow$  (1) of theorem)

$M$  is a splitting field of a

separable polynomial  $f(x) \in K[x]$

then  $M/K$  is Galois

Pf: By induction on  $\deg f(x)$

on the following statement:

Claim

Suppose that  $M/L$  is a finite extension

and that  $h(x) \in L[x]$  is a separable

polynomial &  $\lambda: L \rightarrow M$  is an embedding (not necessarily the identity)

s.t. (1)  $M$  is the splitting field for  $h(x)$



(2)  $M$  is the splitting field  
for  $2(h(x))$

Then  $\exists [M:L]$  embeddings  $\sigma: M \rightarrow M$   
s.t.  $\sigma|_L = \text{id}$ .

Rmk. This statement implies the

Proposition:

- $K = L$
- $\sigma = \text{id}$

- $f(x) = h(x)$

$\Rightarrow \exists [M:K]$  embeddings  $\sigma: M \rightarrow M$   
s.t.  $\sigma|_K = \text{id}$

$$\Rightarrow |\text{Emb}_K(M, M)| = [M:K]$$

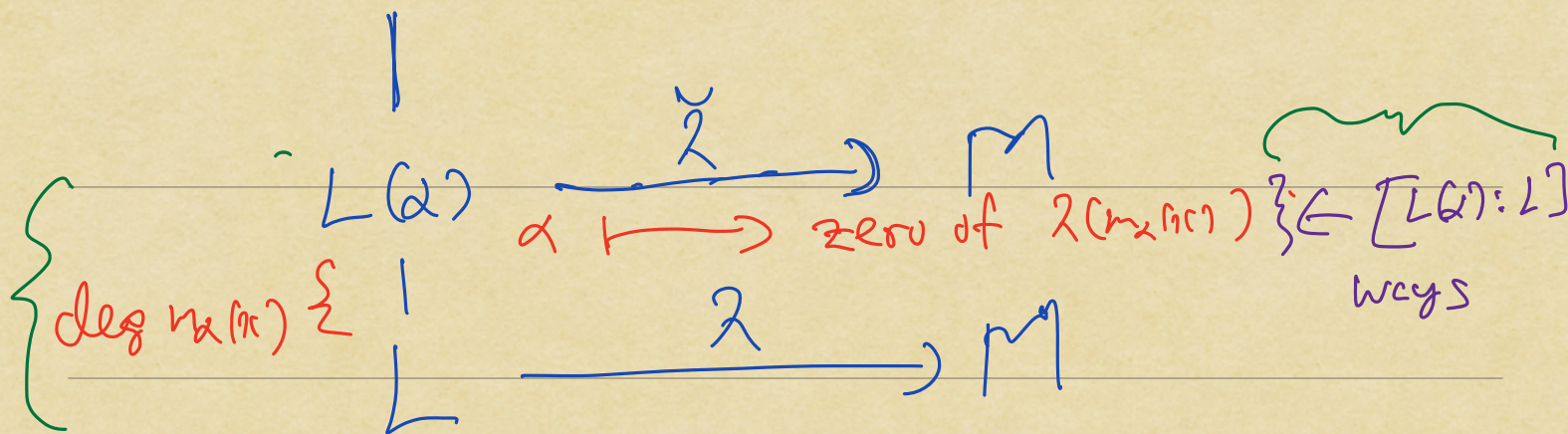
$\Rightarrow M/K$  is Galois.

We will prove the claim by induction  
on  $\deg(h(x))$



Inductive step ← How many?

$$M \xrightarrow{\quad 6 \quad} M$$



$$h(x) \in L[x], \quad \lambda(h(x)) \in M[x]$$

$$\text{Pick } \alpha \in M \text{ s.t. } h(\alpha) = 0$$

$$h(x) \in L(\alpha)[x]$$

$$\parallel$$

$$(x - \alpha) \cdot g(x)$$

$(L(\alpha), g(x), \tilde{\lambda})$  : Apply inductive hypothesis

Note:  $m_\alpha(x) \in L[x] \xRightarrow{h(\alpha)=0} m_\alpha(x) \mid h(x) \in L[x] \Rightarrow \lambda(m_\alpha(x)) \mid \lambda(h(x))$



$\Rightarrow \lambda(\max(\mathcal{M}))$  has all  
its zeros in  $\mathcal{M}$

$\exists [M: L(\alpha)]$  ways of extending  
 $\tilde{\lambda}: L(\alpha) \rightarrow \mathcal{M}$  to

$$\phi: \mathcal{M} \rightarrow \mathcal{M}$$

$\Rightarrow$  In total we have

$[L(\alpha): L]$   $\tilde{\lambda}$  extending  $\lambda$

$[M: L(\alpha)]$   $\phi$  extending  $\tilde{\lambda}$

$\Rightarrow [M: L(\alpha)][L(\alpha): L]$  ways of  
extending  $\lambda$   
||  
 $[M: L]$  to  $\phi$ .



# Galois orbits

$M/K$ : Galois extension

$$\sigma \in G = \text{Gal}(M/K)$$

$$\alpha \in M$$

$$\begin{array}{ccc} M & \xrightarrow{\sigma} & M \\ | & & || \\ K(\alpha) & \xrightarrow{\sigma|_{K(\alpha)}} & M \\ | & & \\ K & & \end{array}$$

$$\sigma|_{K(\alpha)} \in \text{Emb}_K(K(\alpha), M)$$

$\downarrow \cong$

$$\{\beta \in M : m_\alpha(\beta) = 0\}$$

Proposition (2)  $G \curvearrowright M$

An orbit of  $G = \{\beta_1, \dots, \beta_d\}$



Where  $\beta_1, \dots, \beta_d$  have the same minimal polynomial

Rmk: We saw this on the homework in the special case where  $M/K$  is an extension of finite fields

Reformulation:  $f(x) \in K[x]$  irreducible

s.t.  $f(x)$  has a zero in  $M$

$(\iff) f(x)$  has  $d$  distinct zeros

$$\{\beta_1, \dots, \beta_d\} \subseteq M$$

Then  $\{\beta_1, \dots, \beta_d\}$  is a Galois orbit.



Rmk Suppose that  $M/K$  is  
a splitting field of an  
irreducible separable  $f(x) \in K[x]$

$$\{\beta_1, \dots, \beta_d\} \subseteq M \text{ zeros of } f(x) \\ M = K(\beta_1, \dots, \beta_d)$$

Then  $\text{Gal}(M/K) \curvearrowright \{\beta_1, \dots, \beta_d\}$   
acts transitively

$$\Rightarrow \rho: \text{Gal}(M/K) \rightarrow S_d \\ \text{with trivial kernel}$$

We can think of  $\text{Gal}(M/K)$  as  
a subgroup of  $S_d$  acting  
transitively on  $\{1, 2, \dots, d\}$ .



Example:

$$f(x) = \Phi_p(x), \quad d = p-1$$

$$M = (\mathbb{Q}(\zeta_p)) \quad , \quad K = \mathbb{Q},$$

$$\text{Gal}(\mathbb{Q}(\zeta_p)/\mathbb{Q}) \longrightarrow S_{p-1}$$

$\downarrow$

$\parallel$

$$(\mathbb{Z}/p\mathbb{Z})^\times \longrightarrow S_{p-1}$$

$\downarrow$

$$\mathbb{Z}/(p-1)\mathbb{Z}$$

